

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

NAVIGATION CONTROL MODULE (NCM) - ASIA

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Navigation Control Module (NCM) - Japan. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Navigation System](#) (415-00 Information

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B119F-11	GPS Antenna - Circuit short to ground	NOTE: Circuit reference GPS-SIGNAL GPS antenna circuit short circuit to ground	Refer to the electrical circuit diagrams and test the GPS antenna circuit for short circuit to ground
B119F-13	GPS Antenna - Circuit open	NOTE: Circuit reference GPS-SIGNAL GPS antenna circuit open circuit, high resistance	Refer to the electrical circuit diagrams and test the GPS antenna circuit for open circuit, high resistance
B1296-4A	Navigation Map Data Mismatch With Market - Incorrect component installed	- SD memory card map data incorrect to market - Incorrect market set in car configuration file	- Install the correct SD memory card - Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
B12A0-11	Auxiliary Input 2 - Circuit short to ground	NOTE: Circuit reference AUDIO POS / AUDIO NEG Audio positive circuit short circuit to ground	- Refer to the electrical circuit diagrams and test the audio positive circuit for short circuit to ground - Refer to the electrical circuit diagrams and test the audio negative circuit for short circuit to ground

		Audio negative circuit short circuit to ground	
B1300-11	Video Output "A" - Circuit short to ground	NOTE: Circuit reference LVDS IN1 / LVDS IN2 - Low voltage differential signalling positive circuit short circuit to ground - Low voltage differential signalling negative circuit short circuit to ground	- Refer to the electrical circuit diagrams and test the low voltage differential signalling positive circuit for short circuit to ground - Refer to the electrical circuit diagrams and test the low voltage differential signalling negative circuit for short circuit to ground
B1A81-4B	Internal Trip Switch - Over temperature	Navigation control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new navigation control module
B1A84-56	Car Configuration Data - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
U0010-88	Medium Speed CAN Communication Bus - Bus off	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0011-82	Medium Speed CAN Communication Bus Performance -	Invalid data received from another control module via the medium speed CAN	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data

	Alive/sequence counter incorrect / not updated	bus (comfort)	source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-87	Medium Speed CAN Communication Bus Performance - Missing message	Missing message from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Car configuration file mismatch with vehicle specification - Incorrect navigation control module installed	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary - Install a new navigation control module as necessary
U0400-92	Invalid Data Received - Performance or incorrect operation	Touch screen private CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and test the touch screen private CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance